

My work is driven by one question: should you trust a model's predictions? And when a model's internal assumptions are wrong, how can we make its answers verifiable, so they can be trusted anyway? I work across research and engineering, taking a problem from the failure that motivates it through to the system that ships.

EXPERIENCE

Axiom Math

Aug 2026 – Present

Member of Technical Staff · Palo Alto
Building AI mathematicians and verifiable AI: autoformalization and automated theorem proving.

Huawei

2024 – 2026

Staff Engineer · Singapore

Neuro-symbolic mathematical proving (ProofFlow). Lead author on the ICLR 2026 paper and a granted patent. Decomposes a natural-language proof into a DAG of intermediate steps and formalizes each in Lean 4, orchestrating frontier LLMs with cheaper specialized models. Fine-tuned Qwen and DeepSeek with SFT and RL across Nvidia and Huawei Ascend stacks, and generated 100k+ math chain-of-thought traces for foundation-model post-training. Deployed to the Huawei AI assistant, serving 900M+ users.

Geometric design synthesis (PyGeoX). Lead author on the paper and a patent. Teaching LLMs to turn a plain-language description (“a crank 8.7 mm long turning about a fixed centre”) into an exact geometric construction: CAD sketches, linkages, diagrams, every constraint satisfied to numerical precision. Built from scratch a differentiable geometry library (50+ properties, Numba JIT-compiled) scoring each constraint separately, fast enough to verify batched rollouts as a live RL reward; those signals drove SFT/RL on Qwen and Gemma over 100k+ synthetic problems generated with symbolic satisfiability checks. A YOLO + MLLM front-end parses geometry from diagrams. Solves 2.4× more hard problems than standard RL rewards, with an 8B model matching far larger frontier systems.

AI reliability & safety. Training-free, drop-in alignment tooling operating at generation time. Reads internal model signals (hidden states, activations) to detect hallucinated or unsafe content live, then steers the model away via particle filtering and activation steering. Shipped as a unified vLLM and OpenAI-compatible API.

Team. Led 2–4 engineers and researchers across these programs, and supervised three interns.

National University of Singapore

2023 – 2024

Research Fellow

Developed scalable Bayesian nonparametric clustering with uncertainty quantification (R/C++); paired with Vision Transformer features, it reconstructed ancient trade routes from coin imagery at 96% accuracy versus a 55% baseline, featured as a Nature Research Highlight. Statistical and causal-inference consulting for A*STAR and Haleon.

EDUCATION

KAUST

2020 – 2023

Ph.D. in Statistics · Advisor: Håvard Rue

Developed a framework to audit, validate, and robustify hidden assumptions in hierarchical models, with robust non-Gaussian extensions for uncertainty quantification and extreme-event prediction. Open-sourced the [ngvb](#) package (120× speedup over MCMC); results contributed to a \$1M research grant. [AI-Kindi Research Award](#) (top Ph.D. students at KAUST). Led statistics lectures to 45 graduate students.

Instituto Superior Técnico

2014 – 2019

M.Sc. in Applied Mathematics · B.Sc. in Engineering Physics

University of Lisbon, top-ranked in Portugal. Diploma of Academic Merit, Dean's List.

PAPERS AND PROJECTS

Internalizing Geometric Law: Learning from Solver Residuals for Precision-Critical Generation

Cabral, R., Pang, Z., Shou, Z., Shen, X. · [arXiv](#), 2026 · under review at NeurIPS 2026 · [Paper](#) · [GitHub](#)

ProofFlow: A Dependency Graph Approach to Faithful Proof Autoformalization

Cabral, R., Do, T. M., Yu, X., Tai, W. M., Feng, Z., Shen, X. · ICLR 2026 (main conference) · [Paper](#) · [GitHub](#)

Scalable Bayesian clustering for integrative analysis of multi-view data

Cabral, R., de Iorio, M., Harris, A. · 2025 · [Nature Research Highlight](#) · [Application \(Antiquity\)](#) · [Methods \(arXiv\)](#)

Robustness, model checking, and hierarchical models

Cabral, R., Bolin, D., Rue, H. · Journal of the Royal Statistical Society, Series B, 2025 · [Paper](#) · [Package](#)

Fitting latent non-Gaussian models via variational Bayes and Laplace approximations

Cabral, R., Bolin, D., Rue, H. · Journal of the American Statistical Association (Theory & Methods), 2024 · [Paper](#) · [Package](#)

OUTRO

Skills. Python, C/C++, Lean 4, R · PyTorch, Transformers, vLLM, OpenRLHF, LangChain · SFT/RL (LoRA/QLoRA).

Asteroid Discovery. Asteroid [2013 EZ7](#), confirmed and registered by the Minor Planet Center (Harvard & Smithsonian).